Enhancing Habitat Connectivity for Grizzly Bears in Jasper National Park: A Geographic Information System (GIS) Approach

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Introduction & Goal:

Jasper National Park has an incredibly diverse ecosystem, but the grizzly bear stands out as a symbol of wilderness and ecological integrity. As the development of man-made infrastructure within the region accelerates, roads, particularly Highways 16 and 93, carve through the park, disrupting the habitat by creating barriers which limit the natural movements of these wide-ranging animals. These barriers can limit their ability to access food and other environmental necessities and isolate them from others of their species. As a result, the possibility of dangerous wildlife-vehicle collisions increases. To counteract this, the concept of wildlife corridors, particularly overpasses, has emerged as a vital tool in conservation by facilitating a safe passage across these man-made barriers and upholding genetic diversity. This study aims to identify the most advantageous location for a wildlife overpass that would bridge separated habitats, employing GIS-based multi-criteria and least-cost path analyses. The objective is twofold: to maximize connectivity for grizzly bear populations between habitat patches and to do so in an economically viable manner, reflecting the best possible use of conservation resources.

Study Area & Data:

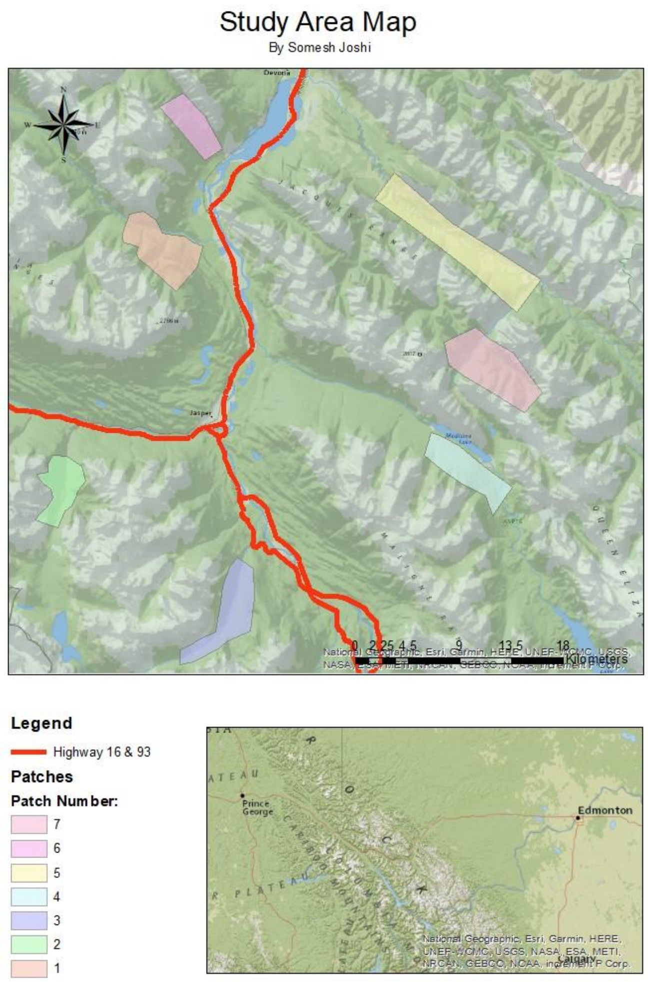

Figure 1

Jasper National Park (Figure 1), a vast expanse of protected wilderness within the Canadian Rockies of Alberta, serves as our study area. Renowned for its rugged topography, the park encompasses over 10,000 square kilometres of a diverse landscape, including glacier-fed lakes, dense coniferous forests, and alpine meadows. The climate is characterized by cold winters and cool summers, supporting a rich biodiversity and a population of grizzly bears that roam across various elevations and ecosystems.

To ensure accuracy in analysis, the data utilized for this study are geo-referenced to a common projection. The raster data have a spatial resolution of 30 meters and include digital elevation models that depict the park's topography alongside vegetation indices which highlight the health and distribution of plant life. These layers also represent the historical extent of wildfires. The raster layers are complemented by vector data which includes the locations and shapes of habitat patches and road networks that are crucial for identifying potential grizzly bear corridors. The data sources are reputable and include the Canadian Digital Elevation Model and Satellite Forest information for Canada.

Methods:

The analysis began by integrating various geospatial datasets to model the landscape's resistance to the movement of grizzly bears. This analytical approach aims to simulate the movement of wildlife through varying habitats and anthropogenic influences with as much accuracy as possible.

Environmental factors relevant to grizzly bear ecology such as elevation, land cover, and proximity to human infrastructure were each represented by raster layers which underlined their distribution across the park. Our goal was to use these layers to measure resistance to movement, a concept extensively discussed in the works of Geneletti (2019) and further supported by ecological studies on bear behaviour (Sawaya et al., 2014). Resistance values were assigned based on literature reviews and expert consultations, ensuring ecological validity. Each layer underwent reclassification to conform to a standardized scale where higher values indicated greater resistance: representing the more challenging areas for bear movement.

After reclassification, a weighted sum process combined the individual resistance layers into a single, cohesive resistance surface. This step was vital as it provided a general blueprint for the cumulative impact of the environmental factors on grizzly bear movement. The weightings were derived from established conservation research (Hilty et al., 2019) and were a direct measure of each layer's relative influence. Human infrastructure and roads were assigned higher values, reflecting their greater impact on wildlife movement.

Once the resistance surface had been established, a least-cost path (LCP) analysis was performed. This technique is widely recognized for its utility in conservation planning (Carroll et al., 2001). The LCP analysis allowed us to identify potential corridors that offered the path of least energetic cost between habitat patches. This essentially mapped out the most accessible routes for the bears by evaluating the path that minimizes the cumulative resistance and, therefore, the potential cost to the bears moving through the landscape.

The identified corridors were then evaluated for their viability, considering not only the cost of traversal for the wildlife but also the potential for integration with existing and planned infrastructure. This ensured the proposed corridors and the subsequent overpass location aligned with both ecological needs and human development plans, allowing for a harmonious approach to landscape management.

Results:

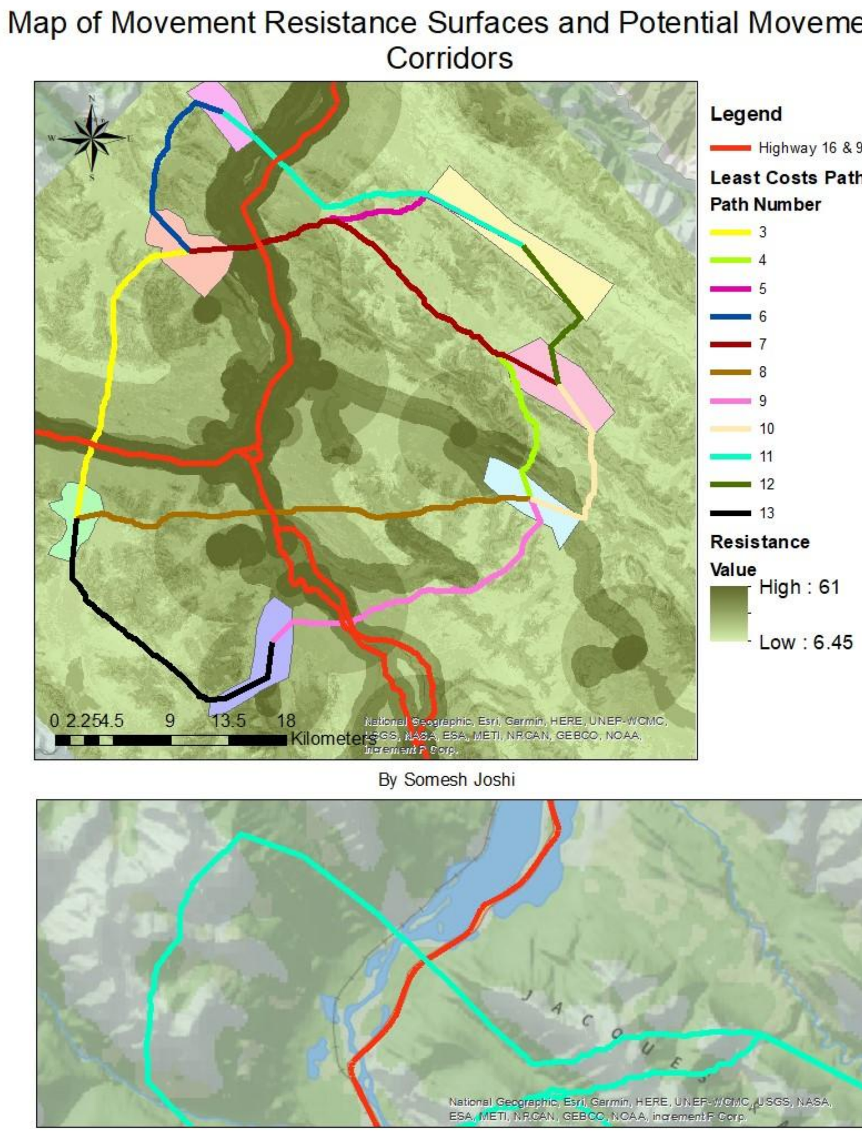

Figure 2

The resistance layer unveiled a pattern where higher resistance zones predominantly aligned with areas of man-made infrastructure and steep terrain (both prominently around roads).

The least-cost path analysis revealed eleven potential movement corridors that connected the seven identified habitat patches. Each corridor was characterized by a distinct path cost, reflecting the cumulative resistance of the route. The corridors varied substantially in their spatial alignment and the habitat patches they connected.

The optimal corridor for overpass placement, based on the least-cost path analysis, was identified as Corridor 11 as highlighted in the inset map (Figure 2). This corridor connected five habitat patches while crossing the human-made barrier, Highway 16. Corridor 11 also had the lowest path cost among corridors intersecting a highway while simultaneously offering the greatest number of connections between habitat patches possible. This provides us with one of the more optimal solutions to connectivity for the investment. The spatial configuration of this corridor, including its intersection points with the highway and the connected habitat patches, is illustrated in the provided map (Figure 2).

Discussion & Conclusion:

The comprehensive GIS analysis and the subsequent identification of Corridor 11 as the preferred location for the wildlife overpass stand as a testament to the power and efficacy of spatial analytical tools in addressing ecological connectivity concerns. The results shed light on the vital need to integrate various environmental factors into conservation planning. This multi-faceted approach also showcases a route that balances ecological benefits against economic costs, in line with Geneletti's (2019) principles of multi-criteria analysis.

The importance of habitat connectivity can be derived as an important insight. Corridor 11 boasts a low cumulative cost and serves its greater purpose of connecting a majority of habitat patches, thereby offering a more holistic solution to wildlife movement constraints imposed by highways. The corridors identified through this study emphasize the delicate interplay between grizzly bear ecology and human infrastructure, echoing findings from previous studies, such as those by Hilty et al. (2019) on the importance of corridor ecology in conservation initiatives.

The application of GIS methodologies allowed for a nuanced analysis, revealing patterns not readily apparent without such detailed landscape evaluation. However, this study is not without its limitations. The static nature of the resistance values used in the model does not fully encapsulate the dynamic nature of wildlife movement as it can vary seasonally and in response to the ever-changing environmental conditions. Furthermore, the assumption of uniform movement behaviour across the bear population may not hold true for bears with unique habitat preferences.

For future research, incorporating real-time tracking data of grizzly bears, if available, could enhance the model's predictive power by providing empirical evidence of actual movement patterns. Additionally, exploring the potential impacts of climate change on habitat quality and connectivity could add another layer of depth to the analysis, ensuring the long-term viability of the proposed corridors.

In conclusion, this study demonstrates the pivotal role that GIS and spatial analysis play in informed decision-making when it comes to wildlife conservation. The identification of Corridor 11 for overpass construction is a strategic recommendation, aimed at ensuring sustainable population dynamics of grizzly bears in Jasper National Park, while also setting a precedent for the application of multi-criteria analysis in environmental planning and management.

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